Eleventh Grade Tile Task

Standard A.APR.2

11

1. a) Remainder of $(x^2 + 4x + 4) \div (x - 1)$ **9**

b) Remainder of $(x^2 + 2x + 0) \div (x - 1)$ **3**

c) Remainder of $(x^2 + 1x + 3) \div (x - 1)$ **5**

2. a) Remainder of $(x^2 + 0x + 1) \div (x - 1)$ **2**

b) Remainder of $(x^2 + 3x + 3) \div (x - 1)$ **7**

c) Remainder of $(x^2 + 1x + 4) \div (x - 1)$ **6**

3. a) Remainder of $(x^2 + 0x + 0) \div (x - 1)$ **1**

b) Remainder of $(x^2 + 3x + 4) \div (x - 1)$ **8**

c) Remainder of $(x^2 + 2x + 1) \div (x - 1)$ **4**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard A.REI.2

11

1. a) Solve x/4 = 2: x = ? **8**

b) Solve √x = 2: x = ? **4**

c) Solve x/9 = 1: x = ? **9**

2. a) Solve x/3 = 1: x = ? **3**

b) Solve x/5 = 1: x = ? **5**

c) Solve √x = 0: x = ? **0**

3. a) Solve √x = 1: x = ? **1**

b) Solve x/2 = 1: x = ? **2**

c) Solve x/7 = 1: x = ? **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard A.REI.7

11

1. a) Solve x + y = 7, x = 4: y = ? **3**

b) Solve x + y = 3, x = 2: y = ? **1**

c) Solve x + y = 9, x = 0: y = ? **9**

2. a) Solve x + y = 3, x = 1: y = ? **2**

b) Solve x + y = 9, x = 3: y = ? **6**

c) Solve x + y = 8, x = 0: y = ? **8**

3. a) Solve x + y = 5, x = 0: y = ? **5**

b) Solve x + y = 6, x = 2: y = ? **4**

c) Solve x + y = 6, x = 6: y = ? **0**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard A.REI.11

11

1. a) x-coordinate where y = 2x meets y = 4 **2**

b) x-coordinate where y = 1x meets y = 9 **9**

c) x-coordinate where y = 1x meets y = 8 **8**

2. a) x-coordinate where y = 1x meets y = 6 **6**

b) x-coordinate where y = 3x meets y = 0 **0**

c) x-coordinate where y = 2x meets y = 6 **3**

3. a) x-coordinate where y = 3x meets y = 3 **1**

b) x-coordinate where y = 1x meets y = 5 **5**

c) x-coordinate where y = 2x meets y = 8 **4**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard A.SSE.3

11

1. a) Smaller root of $x^2 - 17x + 72$ — **8**

b) Smaller root of $x^2 - 9x + 18$ — **3**

c) Smaller root of $x^2 - 15x + 56$ — **7**

2. a) Smaller root of $x^2 - 11x + 18$ — **2**

b) Smaller root of $x^2 - 3x + 2$ — **1**

c) Smaller root of $x^2 - 4x + 0$ — **0**

3. a) Smaller root of $x^2 - 13x + 40$ — **5**

b) Smaller root of $x^2 - 12x + 32$ — **4**

c) Smaller root of $x^2 - 14x + 48$ — **6**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard F.BF.3

11

1. a) $f(x) = x^2 + 4$: value at $x = 0$ — 4

b) $f(x) = x^2 + 0$: value at $x = 0$ — 0

c) $f(x) = x^2 + 2$: value at $x = 0$ — 2

2. a) $f(x) = x^2 + 3$: value at $x = 0$ — 3

b) $f(x) = x^2 + 6$: value at $x = 0$ — 6

c) $f(x) = x^2 + 9$: value at $x = 0$ — 9

3. a) $f(x) = x^2 + 5$: value at $x = 0$ — 5

b) $f(x) = x^2 + 1$: value at $x = 0$ — 1

c) $f(x) = x^2 + 7$: value at $x = 0$ — 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard F.BF.4

11

1. a) $f(x) = x + 1$. Find $f^1(8)$. **7**

b) $f(x) = x + 4$. Find $f^1(6)$. **2**

c) $f(x) = x + 3$. Find $f^1(3)$. **0**

2. a) $f(x) = x + 1$. Find $f^1(6)$. **5**

b) $f(x) = x + 0$. Find $f^1(8)$. **8**

c) $f(x) = x + 0$. Find $f^1(9)$. **9**

3. a) $f(x) = x + 1$. Find $f^1(5)$. **4**

b) $f(x) = x + 0$. Find $f^1(6)$. **6**

c) $f(x) = x + 5$. Find $f^1(8)$. **3**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eleventh Grade Tile Task

Standard F.IF.6

11

1. a) Average rate of change of $y = x^2$ from $x = 0$ to $x = 4$ — **4**

b) Average rate of change of $y = x^2$ from $x = 0$ to $x = 1$ — **1**

c) Average rate of change of $y = x^2$ from $x = 0$ to $x = 2$ — **2**

2. a) Average rate of change of $y = x^2$ from $x = 0$ to $x = 6$ — **6**

b) Average rate of change of $y = x^2$ from $x = 0$ to $x = 9$ — **9**

c) Average rate of change of $y = x^2$ from $x = 0$ to $x = 7$ — **7**

3. a) Average rate of change of $y = x^2$ from $x = 0$ to $x = 8$ — **8**

b) Average rate of change of $y = x^2$ from $x = 0$ to $x = 5$ — **5**

c) Average rate of change of $y = x^2$ from $x = 0$ to $x = 3$ — **3**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard F.IF.7

11

1. **a)** y-intercept of $y = x^2 + 0$ — **0**

   b) y-intercept of $y = x^2 + 8$ — **8**

   c) y-intercept of $y = x^2 + 4$ — **4**

2. **a)** y-intercept of $y = x^2 + 5$ — **5**

   b) y-intercept of $y = x^2 + 2$ — **2**

   c) y-intercept of $y = x^2 + 9$ — **9**

3. **a)** y-intercept of $y = x^2 + 1$ — **1**

   b) y-intercept of $y = x^2 + 6$ — **6**

   c) y-intercept of $y = x^2 + 7$ — **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard F.IF.8

11

1. a) f(x) = x + 6: f(1) **7**

b) f(x) = x + 4: f(1) **5**

c) y = 2^x: value at x = 3 **8**

2. a) f(x) = x + 3: f(1) **4**

b) f(x) = x + 8: f(1) **9**

c) y = 2^x: value at x = 1 **2**

3. a) f(x) = x + 5: f(1) **6**

b) y = 3^x: value at x = 1 **3**

c) y = 3^x: value at x = 0 **1**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.IF.9

11

1. a) Slope of f(x) = 2x + 1 — 2

b) Slope of f(x) = 1x + 0 — 1

c) Slope of f(x) = 9x + 0 — 9

2. a) Slope of f(x) = 7x + 0 — 7

b) Slope of f(x) = 8x + 1 — 8

c) Slope of f(x) = 0x + 0 — 0

3. a) Slope of f(x) = 6x + 1 — 6

b) Slope of f(x) = 5x + 0 — 5

c) Slope of f(x) = 3x + 0 — 3

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.LE.4

11

1. a) log base 10 of 10^2 — 2

b) log base 10 of 10^5 — 5

c) log base 10 of 10^9 — 9

2. a) log base 10 of 10^1 — 1

b) log base 10 of 10^4 — 4

c) log base 10 of 10^8 — 8

3. a) log base 10 of 10^6 — 6

b) log base 10 of 10^3 — 3

c) log base 10 of 10^7 — 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.TF.1

11

1. a) How many full turns in 2π radians? **1**

b) How many full turns in 4π radians? **2**

c) How many full turns in 6π radians? **3**

2. a) How many full turns in 0π radians? **0**

b) How many full turns in 18π radians? **9**

c) How many full turns in 10π radians? **5**

3. a) How many full turns in 12π radians? **6**

b) How many full turns in 8π radians? **4**

c) How many full turns in 16π radians? **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard F.TF.2

11

1. a) Points where the unit circle crosses the x-axis? **2**

b) Arc length on the unit circle for an angle of 1 radians **1**

c) Arc length on the unit circle for an angle of 9 radians **9**

2. a) Arc length on the unit circle for an angle of 0 radians **0**

b) Quadrants of the unit circle? **4**

c) Area of the unit circle (π≈3)? **3**

3. a) Arc length on the unit circle for an angle of 8 radians **8**

b) Arc length on the unit circle for an angle of 5 radians **5**

c) Arc length on the unit circle for an angle of 7 radians **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.TF.5

11

1. a) Amplitude of y = 3 sin x — 3

b) Midline of y = 5 + sin x — 5

c) Midline of y = 0 + sin x — 0

2. a) Amplitude of y = 9 sin x — 9

b) Amplitude of y = 6 sin x — 6

c) Amplitude of y = 4 sin x — 4

3. a) Midline of y = 1 + sin x — 1

b) Amplitude of y = 8 sin x — 8

c) Amplitude of y = 2 sin x — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard N.CN.1

11

1. a) Imaginary part of 1 + 2i — 2

b) Real part of 4 + 1i — 4

c) Real part of 3 + 0i — 3

2. a) Imaginary part of 1 + 9i — 9

b) Imaginary part of 0 + 8i — 8

c) Real part of 7 + 0i — 7

3. a) Real part of 5 + 0i — 5

b) Imaginary part of 1 + 1i — 1

c) Real part of 0 + 0i — 0

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard N.CN.2

11

1. **a)** Real part of (4 + 0i) + (2 + 0i) — 6

   b) Real part of (3 + 0i) + (1 + 1i) — 4

   c) Real part of (0 + 0i) + (0 + 0i) — 0

2. **a)** Real part of (4 + 1i) + (3 + 0i) — 7

   b) Real part of (2 + 1i) + (3 + 0i) — 5

   c) Real part of (1 + 0i) + (0 + 1i) — 1

3. **a)** Real part of (4 + 0i) + (4 + 0i) — 8

   b) Real part of (3 + 1i) + (0 + 1i) — 3

   c) Real part of (1 + 0i) + (1 + 1i) — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard N.CN.7

11

1. a) Real part of the roots of $(x - 5)^2 + 1 = 0$ **5**

b) Real part of the roots of $(x - 9)^2 + 1 = 0$ **9**

c) Real part of the roots of $x^2 + 1 = 0$ $(k = 1)$ **0**

2. a) Real part of the roots of $(x - 1)^2 + 1 = 0$ **1**

b) Real part of the roots of $(x - 3)^2 + 1 = 0$ **3**

c) Real part of the roots of $(x - 4)^2 + 1 = 0$ **4**

3. a) Real part of the roots of $(x - 2)^2 + 1 = 0$ **2**

b) Real part of the roots of $(x - 7)^2 + 1 = 0$ **7**

c) Real part of the roots of $(x - 8)^2 + 1 = 0$ **8**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard S.ID.6

11

1. **a)** Slope of the best-fit line y = 1x + 1 — **1**

   b) Slope of the best-fit line y = 9x + 0 — **9**

   c) Slope of the best-fit line y = 7x + 1 — **7**

2. **a)** Slope of the best-fit line y = 8x + 1 — **8**

   b) Slope of the best-fit line y = 6x + 0 — **6**

   c) Slope of the best-fit line y = 5x + 0 — **5**

3. **a)** Slope of the best-fit line y = 3x + 0 — **3**

   b) Slope of the best-fit line y = 2x + 0 — **2**

   c) Slope of the best-fit line y = 0x + 1 — **0**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard S.ID.7

11

1. a) Interpret the slope: y = 7x + 0, slope? 7

b) Interpret the slope: y = 2x + 0, slope? 2

c) Interpret the slope: y = 0x + 1, slope? 0

2. a) Interpret the slope: y = 8x + 2, slope? 8

b) Interpret the slope: y = 4x + 0, slope? 4

c) Interpret the slope: y = 5x + 2, slope? 5

3. a) Interpret the slope: y = 9x + 0, slope? 9

b) Interpret the slope: y = 3x + 2, slope? 3

c) Interpret the slope: y = 6x + 2, slope? 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE